

SMART INDIA HACKATHON 2026



PROBLEM STATEMENT

Farmers often sell to the only mandi they know, at the only price they're told, to a buyer they can't verify — leading to distress selling and lost income. AgriSphere turns scattered mandi prices into one clear, trustworthy answer for every trade.

Theme : Agriculture, FoodTech & Rural Development

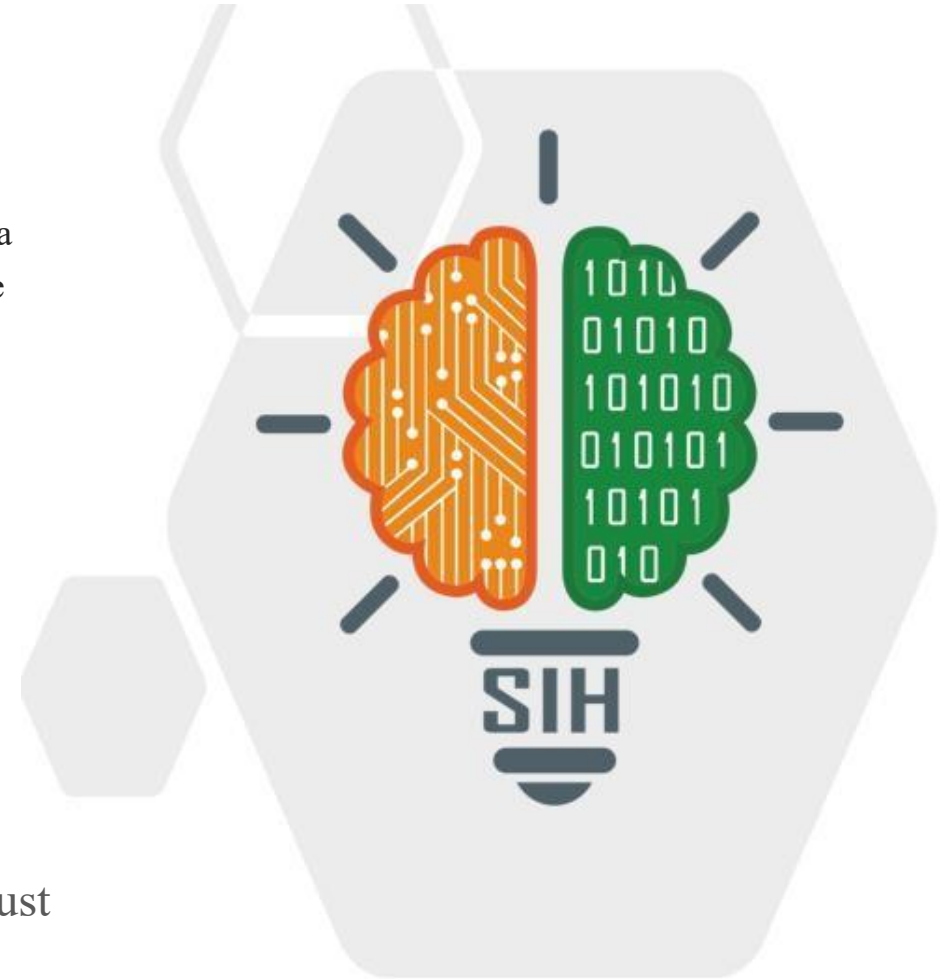
PS Category : Software

Team Name : AgriSphere

Team ID : MMSIH056

AgriSphere

Scattered mandi prices, turned into one clear answer — with trust verified before every deal.



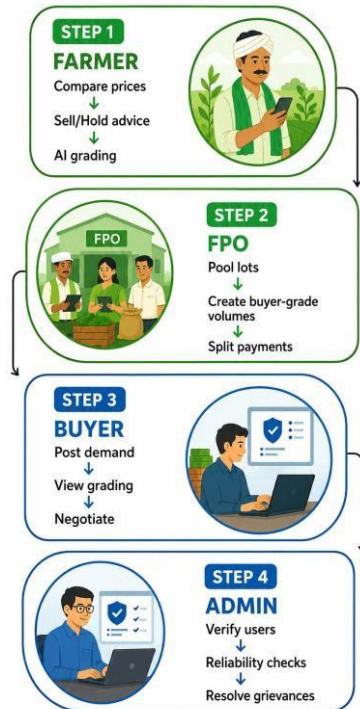
Problem

A farmer sells to the only mandi he knows, at the only price he is told, to a buyer he cannot verify.

Proposed Solution

A single platform that turns scattered mandi prices into one clear answer — **sell now or wait, and to whom** — and lets farmers and buyers check each other's track record before agreeing.

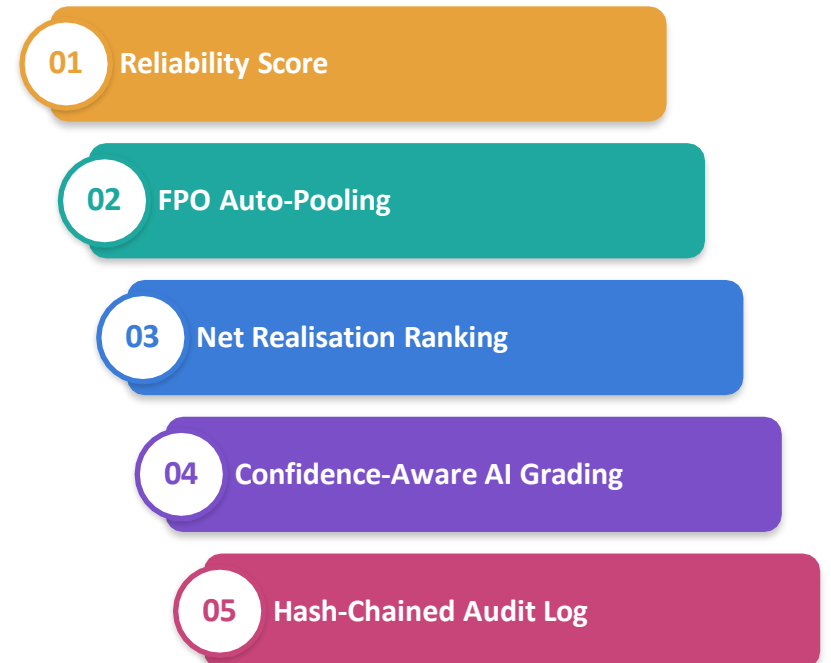
Detailed explanation of the proposed solution



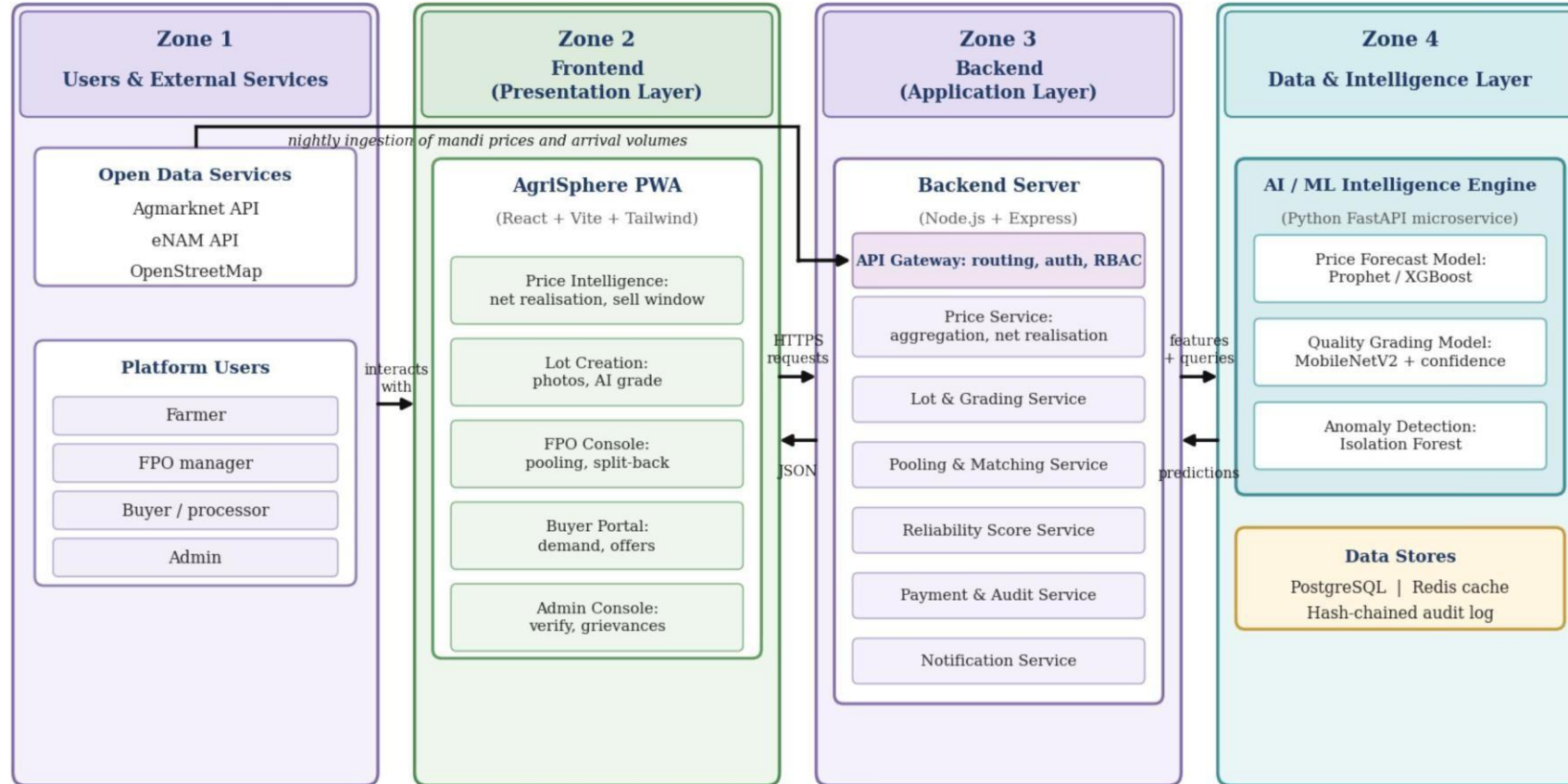
How it addresses the problem



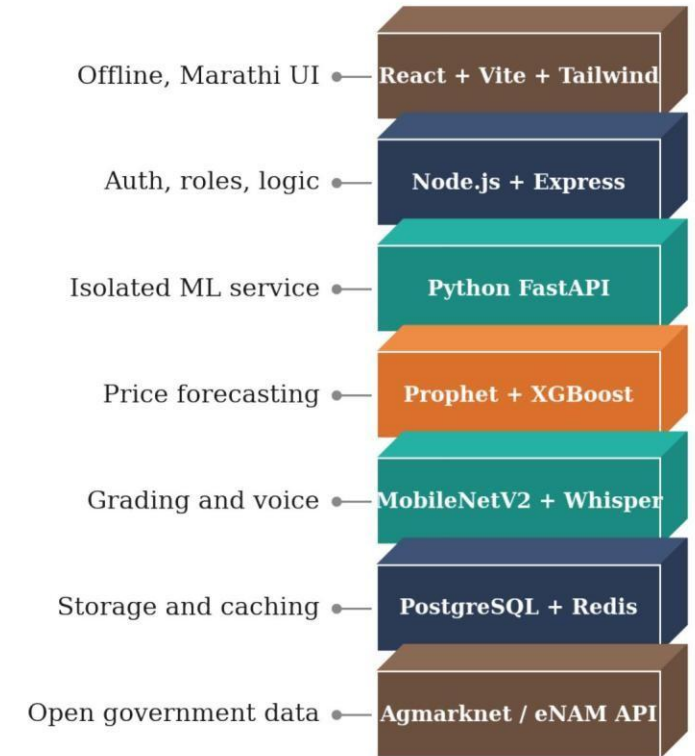
Innovation and uniqueness of the solution



Methodology and process for implementation



Technologies to be used



Feasibility

1

Zero data cost

Agmarknet, eNAM, data.gov.in

2

Ready adoption

FPOs already exist, backed by government

3

Phased build

Finished parts are useful alone

Viability

1

Uses free data and free tools, so costs stay low even as users increase.

2

More transactions improve the reliability score.

3

Can charge a small transaction fee or an FPO subscription.

4

Farmers get better prices, buyers get reliable produce, and FPOs get useful tools.

Challenges and risks, with the strategy for each



Ease of use

Farmers may have limited experience with digital platforms



Simple, familiar interface

Icons, local languages and voice support



Limited Rural Internet

Farmers may have limited experience with digital platforms.



Keep Farmers Connected with SMS

Simple interface with icons, local languages and voice support.



Language gap

Farmers and buyers may use different languages



Localised by design

Structured information displayed in each language



Consignment tracking

Tracking produce during transportation is difficult



Visibility in transit

Clear status updates throughout transportation



Intermediary resistance

Traders may resist greater price transparency



Traders included, not excluded

Verified buyers, not competitors

Potential impact

01

Farmers

Better prices and less distress selling.

02

FPOs

Better aggregation and stronger bargaining power.

03

Buyers & processors

Reliable access to quality produce.

04

Government

Better data for market monitoring and policy decisions.

Benefits of the solution

01

Social

Reduces farmer dependence on traders and strengthens FPO participation and trust.

02

Economic

Helps farmers achieve better prices and reduces selling and transportation costs.

03

Environmental

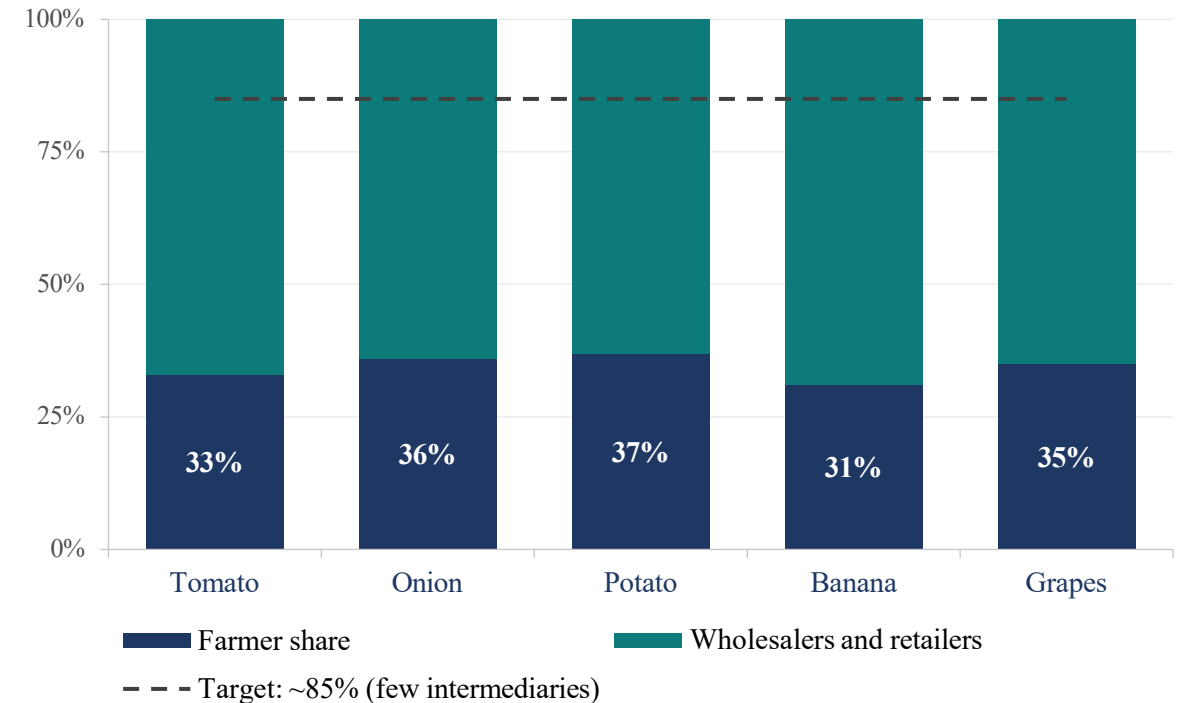
Reduces post-harvest losses and unnecessary travel and fuel use.

04

Technological

Improves grading and price prediction; can expand to dairy, poultry and fisheries.

Where the farmer's money goes today



Sources: RBI working papers on food inflation, 2024; IIM Ahmedabad study on fruit and vegetable marketing



GitHub Repository

<https://github.com/vedantikamahadik2025comp-code/AgriSphere-SIH-2026>

Published Paper

- [1] I. Mahmud, P. R. Das, M. H. Rahman, A. R. Hasan, K. I. Shahin, and D. M. Farid, “Predicting Crop Prices using Machine Learning Algorithms for Sustainable Agriculture,” in Proc. 2024 IEEE Region 10 Symposium (TENSYPMP), New Delhi, India, Sep. 2024. <https://doi.org/10.1109/TENSYPMP61132.2024.10752263>

Data Sources

- [2] Agmarknet, Directorate of Marketing & Inspection, Govt. of India. <https://agmarknet.gov.in>
- [3] eNAM (National Agriculture Market), SFAC, Ministry of Agriculture & Farmers Welfare. <https://enam.gov.in>
- [4] Open Government Data (OGD) Platform India. <https://data.gov.in>

Evidence Base

- [5] R. Roy, S. Gupta, H. Wardhan, S. Sarkar, S. Tewari, R. Bansal, S. Bhatia, and A. Gulati, “Vegetables Inflation in India: A Study of Tomato, Onion and Potato (TOP),” RBI Working Paper WPS (DEPR) 08/2024, Reserve Bank of India, Oct. 2024. <https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=22723>
- [6] R. Das, R. Roy, S. Gupta, S. Bordoloi, R. Kumar, R. Mohan, and A. Gulati, “Price Dynamics and Value Chain of Fruits in India: A Study of Grapes, Bananas and Mangoes,” RBI Working Paper WPS (DEPR) 06/2024, Reserve Bank of India, Oct. 2024. <https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=22721>
- [7] V. P. Gandhi and N. V. Namboodiri, “Marketing of Fruits and Vegetables in India: A Study Covering the Ahmedabad, Chennai and Kolkata Markets,” IIM Ahmedabad Working Paper, 2004. <https://www.iima.ac.in/publication/marketing-fruits-and-vegetables-india-study-covering-ahmedabad-chennai-and-kolkata>
- [8] NABARD Consultancy Services (NABCONS), “Study to Determine Post-Harvest Losses of Agri Produce in India” (ref. year 2020–22), commissioned by Ministry of Food Processing Industries, Govt. of India, 2022. https://www.mofpi.gov.in/sites/default/files/_annex_265_au2145_qhinz4.pdf